

diffblas: algorithmically differentiated BLAS routines

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Summary

diffblas is a library that provides BLAS routines algorithmically differentiated using algorithmic differentiation principles (Griewank & Walther, 2008) from their reference implementations in LAPACK on GitHub using the automatic differentiation tool Tapenade (Hascoët & Pascual, 2013). It supports four modes: forward (_d), vector forward (_dv), reverse (_b), and vector reverse (_bv).

In addition to differentiating the standard Fortran-style BLAS interface, diffblas also provides differentiated CBLAS routines, facilitating interoperability with C and other languages. Its API mirrors BLAS / CBLAS, with additional arguments specifying differentiation variables, making integration into existing workflows straightforward.

diffblas calls the underlying standard BLAS implementation, and is agnostic to the backend (OpenBLAS, BLIS, MKL, Apple Accelerate), ensuring both performance and portability. Precompiled artifacts are available on GitHub to simplify installation and usage across different programming environments.

By providing efficient and accurate derivatives of linear algebra operations, diffblas facilitates gradient-based optimization, sensitivity analysis, and derivative-based workflows in scientific computing.

Statement of need

Linear algebra routines such as those in LAPACK are widely used in scientific computing, optimization, and machine learning. However, they do not provide derivatives, which are often required for gradient-based algorithms.

Existing approaches rely on hand-coded derivatives or generic automatic differentiation applied to high-level code, which can be inefficient or error-prone (Jonasson et al., 2020).

diffblas addresses this gap by providing algorithmically differentiated BLAS routines directly from reference LAPACK implementations and following relevant differentiation rules (Giles, 2008).

State of the field

Automatic source-to-source differentiation tools, such as Tapenade (Hascoët & Pascual, 2013), ADOL-C (Griewank et al., 1996), or TAF (Giering & Kaminski, 2003), provide general mechanisms to compute derivatives of code. However, naively applying AD to low-level BLAS or LAPACK routines can be inefficient due to loop structures, memory layout, and caching constraints (Jonasson et al., 2020). diffblas addresses these challenges by

38 generating differentiated routines directly from reference LAPACK implementations, combining
39 the reliability and performance of LAPACK with the efficiency of algorithmic differentiation.

40 Research impact statement

41 This work was inspired in part by a need to differentiate a Fortran code (Hascoët et al.,
42 2026) that uses BLAS and LAPACK routines, and to use the differentiated application for
43 gradient-based optimization.

44 Providing both the standard and CBLAS interfaces ensures that diffblas can be adopted across
45 different programming environments, facilitating derivative computations in diverse scientific
46 computing projects. Precompiled artifacts on GitHub further simplify integration, enabling
47 rapid deployment in multiple languages and scientific computing projects.

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53 AI usage disclosure

54 Generative AI was used to ... ChatGPT was used to check spelling, grammar, and clarity of the
55 English text in this paper.

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